

The measurable strength of cofinal-orbit rigidity

An equiconsistency theorem and a canonical weakly homogeneous forcing decomposition

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18 September 2026

Abstract. The supplied synthesis proves that ultraexacting cardinals admit rigid classes of cofinal ω -sequences modulo finite difference. We determine the consistency strength of this particular conclusion: the existence of one such class is equiconsistent with one measurable cardinal. The same upper bound supplies λ distinct rigid classes with pairwise disjoint representatives, simultaneously witnessing the full-width obstruction to definable complete sections. The upper bound is an ordinary Prikry extension. Its essential additional feature is that the cone isomorphisms preserve *the entire finite-change class*, although they do not preserve its representative. The lower bound is an application of the classical Dodd–Jensen covering theorem.

We also give a forcing interpretation of each rigid class q . For every $a \in q$, $\text{HOD}_{\{a\}}$ is a proper set-forcing extension of $\text{HOD}_{\{q\}}$ by a canonically coded, atomless, weakly homogeneous Boolean algebra. Every representative gives a generic filter for the same algebra and the same extension. Under the additional hypothesis $V_\lambda \subseteq \text{HOD}$, this extension adds no bounded subsets of λ but changes its cofinality from λ to ω . The arguments distinguish an orbit used as one parameter from a representative used as a parameter throughout.

Main conclusion

Writing RC for the rigid-class assertion defined precisely in Section 1, we prove

$$\text{Con}(\text{ZFC} + \exists \kappa \text{ Meas}(\kappa)) \iff \text{Con}(\text{ZFC} + \text{RC}).$$

This calibrates a consequence of ultraexactingness, not ultraexactingness itself. It shows that the rigid-class package, including its full-width consequences, already occurs at measurable-cardinal consistency strength. The new contribution is the calibration and forcing analysis of the dossier’s exact definition of rigidity; the underlying Prikry, Vopěnka, and Dodd–Jensen machinery is classical.

Research status. The new deductions are proved below in English. The deep covering theorem is an explicitly identified input, not a claimed new proof of fine structure. No global priority claim or Lean verification is asserted.

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1 Statements and relation to the supplied research

The reference point is the second-edition *Large Cardinals Synthesis* in Cardinals3.zip, especially its subsection “Rigid classes” and the questions following its analysis of the cofinal quotient [1]. We retain its definition of rigidity. We do not assume any of its proposed consistency constructions in proving the new upper bound.

1.1 Parameters, cofinal sets, and rigidity

All unqualified cardinalities and all occurrences of OD are computed in the ambient universe $V \models \text{ZFC}$. For a collection C of allowed parameters, OD_C permits ordinal parameters and finitely many elements of C . HOD_C is the corresponding hereditary class. In particular, in $\text{OD}_{V_\lambda \cup \{q\}}$, the set q is *one additional parameter*; its members are not thereby allowed as parameters. The distinction is essential. We abbreviate $\text{OD}_{\{q\}}$ to OD_q when no ambiguity is possible.

For an uncountable cardinal λ of cofinality ω , put

$$\begin{aligned} D_\lambda &= \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}, \\ C_\lambda &= \{a \subseteq \lambda : \sup a = \lambda, |a| < \lambda\}, \\ a =^* b &\iff |a \triangle b| < \omega, \\ Q_\lambda &= D_\lambda / =^*. \end{aligned}$$

An element of Q_λ is a set, despite the word “class.” For $a \in D_\lambda$,

$$[a] = \{a \triangle d : d \in [\lambda]^{<\omega}\}.$$

Every finite modification of a is still a cofinal set of order type ω . There are exactly λ such modifications: the displayed parametrization gives the upper bound, and $\alpha \mapsto a \triangle \{\alpha\}$ gives the lower bound.

Definition 1.1 (The dossier’s rigidity property). For $q \in Q_\lambda$, let $\text{Rig}(\lambda, q)$ assert that

$$\emptyset \neq A \subseteq C_\lambda, \quad A \in \text{OD}_{V_\lambda \cup \{q\}} \implies |A| \geq \lambda.$$

The quantified family A need not consist of representatives of q . It can be any definable family of short cofinal subsets of λ .

We will use three different hereditary classes:

$$N_q = \text{HOD}_{V_\lambda \cup \{q\}}, \quad H_q = \text{HOD}_{\{q\}}, \quad M_a = \text{HOD}_{\{a\}} \quad (a \in q). \quad (1.1)$$

The first is an inner model of ZF and contains V_λ ; it is not assumed to satisfy Choice. The latter two are inner models of ZFC, since a single fixed parameter gives a canonical definable well-order of its hereditarily definable sets. Neither this fact nor the notation implies $q \in H_q$. Indeed, for a rigid class $q \notin H_q$.

These uses of definability are first-order expressible by the usual rank-local coding of ordinal definitions: use a formula, a rank V_θ in which it is evaluated, and finitely many ordinal parameters. They do not require a truth predicate for the entire universe.

1.2 The theories being calibrated

Definition 1.2. Let $RS(\lambda)$ be the conjunction

$$\lambda > \omega \text{ is a cardinal,} \quad \text{cf}(\lambda) = \omega, \quad (\forall \alpha < \lambda) |V_\alpha| < \lambda.$$

This implies that λ is strong limit and $|V_\lambda| = \lambda$. Define the following first-order assertions over ZFC.

HR: there is a λ satisfying $RS(\lambda)$ which is regular in HOD_{V_λ} .

RC: there are λ and $q \in Q_\lambda$ such that $RS(\lambda)$ and $\text{Rig}(\lambda, q)$.

MR: there are a λ satisfying $RS(\lambda)$ and a sequence $\langle a_i : i < \lambda \rangle$ of pairwise disjoint members of D_λ such that every $q_i = [a_i]$ is rigid and all the classes N_{q_i} are equal.

The rank bound in RS matches the setting of the supplied reports. It is stronger than merely saying “strong limit of countable cofinality.” The uncountability requirement is also indispensable: at $\lambda = \omega$ the family C_λ is empty and the rigidity definition would be vacuous.

Theorem 1.3 (Consistency calibration). The following theories are equiconsistent:

$$\text{ZFC} + \exists \kappa \text{ Meas}(\kappa), \quad \text{ZFC} + \text{HR}, \quad \text{ZFC} + \text{RC}, \quad \text{ZFC} + \text{MR}.$$

More specifically, $\text{MR} \Rightarrow \text{RC} \Rightarrow \text{HR}$ in ZFC; HR implies the existence of an inner model with a measurable cardinal; and ordinary Prikry forcing over a measurable cardinal forces MR.

The proof occupies Sections 2–6. Sections 7–8 prove the complementary forcing structure theorem. There is no assertion that the measurable cardinal from the lower bound is measurable in V , or that it must be the ambient singular cardinal λ .

1.3 What is extended, and what is imported

Ingredient	Role in this continuation
Rigidity from ultraexactingness	Prior dossier result; motivation only, not an input to the new measurable upper bound.
Rigidity versus regularity	An exact characterization, proved in Section 2.
Prikry’s theorem	Classical forcing machinery; the relevant proof is included. The additional parameter bookkeeping preserves the entire class q .
Dodd–Jensen covering	Deep classical input used only for the lower bound. Its application is proved explicitly.
Vopěnka’s algebra	Classical construction; specialized and proved here, including the common-extension and finite-change homogeneity conclusions.
Many copies and saturation	The underlying deductions already occur in the dossier. Here they are realized from a single measurable cardinal, rather than an ultraexacting hypothesis.

A related current result should be distinguished from this calibration. In a preprint posted on 25 August 2026, Benhamou and coauthors [9] obtain the optimal measurable-cardinal bound for making $\aleph_\omega$ the first ordinal at which the ambient and HOD power sets disagree. Their statement concerns a first-disagreement phenomenon, not the exact rigid quotient class with all V_λ parameters used here. It is not an input to our proof. The present contribution is a continuation of the supplied dossier; neither this comparison nor the literature check establishes global priority for the formulation below.

2 Rigidity is exactly relative HOD regularity

The first observation removes the quantification over families from the *test* for rigidity, without weakening the property.

Lemma 2.1 (Least-order-type compression). Suppose λ is an infinite cardinal and $\emptyset \neq A \subseteq C_\lambda$ has $|A| < \lambda$. There is a cofinal $u \subseteq \lambda$ with $|u| < \lambda$ which is definable from A and λ , without choosing a member of A .

Proof. Let τ be the least ordinal occurring as the order type of a member of A , and let

$$A_\tau = \{a \in A : \text{ot}(a) = \tau\}, \quad u = \bigcup A_\tau.$$

Since the members of A have cardinality below the initial ordinal λ , their order types are also below λ . Thus $\tau < \lambda$. The family A_τ is nonempty, so u is cofinal. In the ambient universe,

$$|u| \leq |A_\tau| \cdot |\tau| < \lambda.$$

The last inequality uses a product of two *fixed* cardinals below λ . It does not assume that λ is regular. All the operations used to define u are canonical from A . $\square$

Lemma 2.2 (Order type detects external smallness). If $B \subseteq \lambda$ and $|B|^V < \lambda$, where λ is a cardinal of V , then $\text{ot}(B) < \lambda$. If B belongs to a transitive inner model, its order type and increasing enumeration are absolute between that model and V .

Proof. The increasing enumeration is an order-isomorphism with a unique ordinal. If that ordinal were at least the initial ordinal λ , it would give an injection of λ into B , contrary to the assumed cardinality. The well-founded recursion computing the enumeration is absolute between transitive models with the same ordinals. $\square$

Theorem 2.3 (Intrinsic regularity characterization). For every uncountable cardinal λ of cofinality ω and every $q \in Q_\lambda$,

$$\text{Rig}(\lambda, q) \iff N_q \models \text{“}\lambda \text{ is regular.”}$$

Equivalently, there is no cofinal $u \subseteq \lambda$ of order type below λ in $\text{OD}_{V_\lambda \cup \{q\}}$.

Proof. If N_q contains a cofinal map $f : \tau \rightarrow \lambda$ with $\tau < \lambda$, then its range u is a short cofinal set in N_q . In particular, $u \in \text{OD}_{V_\lambda \cup \{q\}}$, and the definable singleton $\{u\}$ violates rigidity. Notice that an ordinal domain below λ is small in V , even if the two models disagree about some other cardinalities.

Conversely, suppose that rigidity fails, witnessed by a nonempty definable $A \subseteq C_\lambda$ of cardinality below λ . Apply Lemma 2.1 to obtain a definable short cofinal set u . A definable set of ordinals is hereditarily definable from the same allowed parameters, because every ordinal is ordinal definable. Thus $u \in N_q$. By Lemma 2.2, its increasing enumeration has an ordinal domain $\tau < \lambda$, and the same enumeration belongs to N_q . This contradicts regularity there.

Finally, a cofinal set of order type below λ gives a singleton counterexample, whereas compression produces such a set from any counterexample family. This proves the equivalent formulation as well. $\square$

Corollary 2.4. If q is rigid, then λ is regular in H_q and in HOD_{V_λ} . Moreover $q \cap N_q = \emptyset$ and $q \notin N_q$.

Proof. Both smaller hereditary classes are contained in N_q , so a cofinal map in either would be one in N_q . A member of q in N_q would itself be a cofinal set of order type $\omega < \lambda$. Finally, transitivity would put every member of q in N_q if q belonged to N_q . $\square$

This characterization is useful but not a solution of every classification question in the dossier: it does not decide which classes are rigid from finite combinatorial data, nor does it produce distinct hereditary models from distinct rigid classes.

3 The measurable lower bound

Here is the only deep inner-model input to the calibration.

Input A (Dodd–Jensen covering below a measurable). If there is no inner model of ZFC with a measurable cardinal, the Dodd–Jensen core model K has a canonical ordinal-definable well-order and covers every uncountable set X of ordinals by a set $Y \in K$ with $X \subseteq Y$ and $|Y|^V = |X|^V$.

We use the classical covering theorem, not a conjectural covering principle for HOD. The original sources are Dodd–Jensen [2, 3]; Mitchell explicitly records the resulting singular-cardinal dichotomy in his introduction [4]. The proof of the covering theorem itself requires core-model fine structure and is not reproduced here.

Proposition 3.1 (No inner measurable, no HOD regularization). If an uncountable singular cardinal λ of V is regular in HOD, then there is an inner model with a measurable cardinal. Consequently HR, RC, and MR each imply such an inner model.

Proof. Assume that there is no inner model with a measurable cardinal, and use Input A. Choose a cofinal $c \subseteq \lambda$ with $|c| = \text{cf}(\lambda) < \lambda$. An uncountable singular cardinal is a limit cardinal, so $\omega_1 < \lambda$. Therefore

$$X = c \cup \omega_1$$

is an uncountable set of ordinals of cardinality below λ . Cover it by $Y \in K$ of the same ambient cardinality, and put $B = Y \cap \lambda$. Then $B \in K$, B is cofinal in λ , and $|B|^V < \lambda$.

The order type $\tau = \text{ot}(B)$ is below λ by Lemma 2.2. Hence K contains an increasing cofinal map $\tau \rightarrow \lambda$. Every set of K is ordinal definable in V , by its position in the canonical well-order of K ; the same is true of its transitive closure. Thus $K \subseteq \text{HOD}$, and the cofinal map belongs to HOD. This contradicts the stipulated regularity of λ there.

For the final assertions, regularity in HOD_{V_λ} implies regularity in its submodel HOD. Corollary 2.4 applies to RC, and MR includes an instance of RC. $\square$

Corollary 3.2 (Conditional inconsistency). The conjunction of ZFC, “there is no inner model with a measurable cardinal,” and “there is a rigid class at an uncountable singular cardinal of cofinality ω ” is inconsistent.

This is a lower bound, not an assertion that a singular cardinal becomes measurable in the ambient universe. Nor does it use the large-cardinal embedding assumptions from which the supplied dossier originally obtained rigidity.

4 A quotient-preserving homogeneity principle

Ordinary homogeneity is insufficient as a slogan here: we must specify *which name* survives the cone isomorphisms.

Definition 4.1. Let $W \models \text{ZFC}$, let $\mathbb{P} \in W$ be a set forcing, and let $\dot{q}$ be a $\mathbb{P}$ -name. Say that the forcing is *cone homogeneous over $\dot{q}$* if, for any $p, r \in \mathbb{P}$, there are $p' \leq p, r' \leq r$ and an isomorphism between the cones below p' and r' which preserves $\dot{q}$ and the check names for ground-model parameters. Preservation of $\dot{q}$ means equality of its transported and original interpretations under the corresponding generic filters.

Lemma 4.2 (The fixed-name decision lemma). Under Definition 4.1, every sentence $\varphi(\dot{q}, \check{x}_1, \dots, \check{x}_m)$, with $x_i \in W$, is decided by the top condition. In $V = W[G]$, every set of ordinals definable from $q = \dot{q}^G$ and finitely many ground-model parameters belongs to W .

Proof. Suppose two conditions force opposite values of the sentence. Strengthen them to the isomorphic cones given by the hypothesis. The forcing isomorphism theorem transports the sentence to itself, so the first cone would force both a sentence and its negation, a contradiction. Since the conditions deciding any sentence are dense, the top condition decides it.

Now let $A \subseteq \beta$ in the extension be defined from q and ground parameters by ψ . For every $\xi < \beta$, apply the first part to the sentence expressing membership of ξ in the uniquely defined set. The forcing relation is a definable relation of W , and therefore

$$A = \{\xi < \beta : \mathbf{1} \Vdash \text{“}\xi \text{ belongs to that set”}\}$$

is a set in W . The ordinal β may be included among the parameters. This also handles definitions originally presented as unique-object definitions rather than as membership formulas. $\square$

Theorem 4.3 (Rigidity transfer). Suppose $V = W[G]$ satisfies all of the following: λ is regular in W ; it remains an uncountable cardinal and has cofinality ω in V ; $V_\lambda^V = V_\lambda^W$; $q = \dot{q}^G \in Q_\lambda^V$; and $\mathbb{P}$ is cone homogeneous over $\dot{q}$. Then $\text{Rig}^V(\lambda, q)$.

Proof. Assume that A is a counterexample to rigidity in V . By Lemma 2.1, there is a cofinal $u \subseteq \lambda$, of ambient cardinality below λ , definable from q , ordinal parameters, and finitely many elements of V_λ^V . All the latter parameters are in W . Lemma 4.2 therefore gives $u \in W$. But $\text{ot}(u) < \lambda$ in V , and its order type is absolute to W . Its increasing enumeration contradicts regularity of λ in W . $\square$

Remark 4.4. No claim is made that the individual representative of q is fixed by the isomorphisms. That would be incompatible with the intended singularization. The protected information is precisely its finite-change class.

5 The Prikry construction

Let $W \models \text{ZFC}$ contain a measurable cardinal κ , and fix a normal nonprincipal κ -complete ultrafilter U on κ in W . All forcing constructions in this section are performed in W . We recall enough of the ordinary Prikry argument to account for every hypothesis of Theorem 4.3. The classical source is Prikry [5]; a modern research treatment with detailed proofs is Benhamou [6].

5.1 Conditions and the Prikry property

A condition in $\mathbb{P}_U$ is a pair (s, A) , where s is a finite increasing sequence from κ , $A \in U$, and every member of A is larger than $\max(s)$ when $s \neq \emptyset$. A stronger condition $(t, B) \leq (s, A)$ has s as an initial segment of t , adds to the stem only points of A , and has $B \subseteq A$. A *direct* extension has the same stem. We identify finite increasing sequences with their ranges when taking unions or symmetric differences.

Normality has the following two familiar consequences. We include their short derivations to make the subsequent fusion argument precise.

Lemma 5.1 (Diagonalization and finite homogeneity). For a family $\langle A_t : t \in [\kappa]^{<\omega} \rangle$ of members of U , there is $B \in U$ such that

$$t \in [B]^{<\omega}, \beta \in B, \max(t) < \beta \implies \beta \in A_t.$$

The empty-stem requirement can be imposed by intersecting with $A_\emptyset$. Also, if $h : [\kappa]^{<\omega} \rightarrow r$ with r finite, there is $B \in U$ on which h is constant separately on each $[B]^n$, $n < \omega$.

Proof. For each $\alpha < \kappa$, intersect the A_t for $t \subseteq \alpha$. There are fewer than κ such finite sets, so the intersection is in U . Take the normal diagonal intersection of these intersections and restrict it to limit ordinals. Given $\max(t) < \beta$ in this diagonal set, choose α strictly between $\max(t)$ and β ; the desired membership follows. For empty t , make the indicated extra intersection. Normal measures contain all clubs, including the set of limit ordinals.

For the second assertion, first prove homogeneity on n -element sets by induction on n . The case $n = 1$ is the ultrafilter property. For a coloring of $(n + 1)$ -element sets, and each n -element stem t , choose the unique color taken on a U -large set of possible last points. Diagonalize these large sets by the first assertion. Apply the inductive hypothesis to the resulting coloring of n -element stems, and intersect with the diagonal set. This makes the original coloring constant on $(n + 1)$ -element sets. Finally intersect the homogeneous sets obtained for all finite n . Countable completeness preserves membership in U . $\square$

Lemma 5.2 (Prikry property). For any forcing sentence φ and any $(s, A) \in \mathbb{P}_U$, a direct extension of (s, A) decides φ .

Proof. For each finite extension t of the stem s , assign a color $h(t)$: use 1 if some condition with stem t forces φ , use -1 if some condition with that stem forces $\neg\varphi$, and use 0 if no condition with that stem decides the sentence. The first two alternatives cannot both occur, because two conditions with the same stem are compatible by intersection of their measure-one sets. In a nonzero case, choose a witnessing tail set $A_t \in U$; in a zero-color case put $A_t = \kappa$.

Apply Lemma 5.1 to the coloring of finite continuations u given by $h(s \smallfrown u)$. Then diagonalize the chosen tail sets, retaining a measure-one subset $B \subseteq A$ such that the tail of B after u is included in $A_{s \smallfrown u}$. The requirements for the fixed stem s and for the empty continuation are met by an additional intersection. On B , the color of $s \smallfrown u$ depends only on the length of u .

Some extension of (s, B) decides φ , by density. Let n be its number of additional stem points and let $\varepsilon \in \{1, -1\}$ be its color. For every $v \in [B]^n$, the condition

$$(s \smallfrown v, B \setminus (\max(v) + 1))$$

forces the same value ε , using the diagonal refinement. For $n = 0$, read this as the condition with stem s itself. Every condition below (s, B) can be extended to have at least n additional stem points; it then extends one of these deciding conditions. Thus a dense set below (s, B) forces the same value, and (s, B) decides φ . $\square$

5.2 The extension preserves the entire lower rank

Proposition 5.3. If $G \subseteq \mathbb{P}_U$ is generic over W and $V = W[G]$, then all cardinals are preserved, $\text{cf}^V(\kappa) = \omega$, no bounded subsets of κ are added, and

$$V_\kappa^V = V_\kappa^W, \quad (\forall \alpha < \kappa) |V_\alpha|^V < \kappa.$$

Proof. The union c of the stems in G is an increasing sequence of length ω . Conditions with arbitrarily long stems are dense, and for each $\beta < \kappa$ the conditions placing a stem point above β are dense. Hence c is cofinal in κ .

Let $\mu < \kappa$, and let $\dot{x}$ be a name for a subset of μ . Starting from any condition, decide the statements $\xi \in \dot{x}$, $\xi < \mu$, successively by direct extensions using Lemma 5.2. At limits, intersect the tail sets. κ -completeness supplies a single direct extension deciding the entire subset. Therefore every subset of μ in V already belongs to W . In particular, every cardinal below κ is preserved, since a bijection witnessing its collapse could be coded by a bounded subset of κ .

The cardinal κ is inaccessible in W . Its smaller cardinals remain cardinals and are unbounded in κ , so κ cannot be collapsed: a purported injection into a smaller ordinal would already collapse some preserved cardinal below κ . Finally, there are only κ finite stems, and conditions with the same stem are compatible. Thus $\mathbb{P}_U$ has the κ^+ -chain condition and preserves all cardinals above κ as well.

For the rank assertion, argue by induction on $\alpha < \kappa$. In W , every V_α has cardinality below κ . Once V_α has been shown unchanged, a ground-model bijection with a smaller ordinal codes every subset of V_α by a bounded subset of κ . Consequently its power set has not changed either. Limit stages follow by taking unions. The ground-model bijections witnessing $|V_\alpha| < \kappa$ are still available in the extension. $\square$

5.3 The tail class, unlike the sequence, is preserved

Let $\dot{c}$ be the canonical generic cofinal set and define the name

$$\dot{q} = \{\dot{c} \triangle \check{d} : d \in [\kappa]^{<\omega}\}.$$

In every generic extension, $\dot{q}$ is exactly $[\dot{c}]$.

Lemma 5.4 (Prikry cones preserve the finite-change class). $\mathbb{P}_U$ is cone homogeneous over $\dot{q}$.

Proof. Take $p = (s, A)$ and $r = (t, B)$. Choose

$$C \in U, \quad C \subseteq A \cap B, \quad \min C > \max(s \cup t),$$

with the evident interpretation for empty stems. Put $p' = (s, C)$ and $r' = (t, C)$. The map

$$(s \smallfrown u, D) \longmapsto (t \smallfrown u, D)$$

is an order isomorphism between the two cones. The same finite continuation u and tail set D are permitted on either side. The stems s and t need not have equal lengths.

Corresponding generics have cofinal sets $s \cup h$ and $t \cup h$ for a common tail h . These differ by the finite set $s \triangle t$, so their finite-change classes are equal. Moreover each cofinal set is definable from the other and the two finite ground-model stems; the resulting forcing extensions are the same. Thus the transported name for q has exactly the required interpretation, and check names for ground-model parameters are unchanged. $\square$

Theorem 5.5 (A rigid class from one measurable). In every ordinary Prikry extension $V = W[G]$ as above,

$$\text{RS}(\kappa) \quad \text{and} \quad \text{Rig}^V(\kappa, [c])$$

hold. In particular, $\text{HOD}_{V_\kappa \cup \{[c]\}}^V$ regards κ as regular.

Proof. Proposition 5.3 gives $\text{RS}(\kappa)$ and all rank and cardinality hypotheses of Theorem 4.3. Lemma 5.4 gives the remaining homogeneity hypothesis. Apply the transfer theorem, and then Theorem 2.3. $\square$

Remark 5.6 (Uniformity over low-rank parameters). This conclusion is not restricted to parameters that were ordinal definable in W . Every member of the entire ambient V_κ is a ground-model parameter, and the cone isomorphisms preserve all their check names. This is why the construction proves the dossier's full $\text{OD}_{V_\kappa \cup \{q\}}$ version of rigidity.

6 Multiplicity, saturation, and the completed calibration

6.1 One class supplies many copies without additional strength

Lemma 6.1 (Disjoint mutually definable copies). For every $q \in Q_\lambda$ there are λ distinct classes $q_i \in Q_\lambda$, $i < \lambda$, mutually ordinal definable with q , having pairwise disjoint representatives. Thus if q is rigid, every q_i is rigid and $N_{q_i} = N_q$.

Proof. Use the canonical pairing $\pi : \lambda \times \lambda \rightarrow \lambda$ obtained by ordering pairs first by their maximum coordinate and then lexicographically. This well-order has order type λ : every pair has fewer than λ predecessors, and there are λ pairs. For each $i < \lambda$, the map $f_i(\alpha) = \pi(i, \alpha)$ is strictly increasing and cofinal in λ . Its range is disjoint from the ranges of the other f_j .

Choose any $a \in q$ and put $a_i = f_i''a$, $q_i = [a_i]$. These representatives are pairwise disjoint cofinal ω -sets. The definition of q_i does not depend on the chosen a : replacing a by a finite modification only finitely modifies $f_i''a$. Thus q_i is definable from q and i .

Conversely, for any $b \in q_i$, pull $b \cap \text{ran}(f_i)$ back along f_i . The resulting set differs finitely from a , regardless of b . Its equivalence class is therefore q , giving a definition of q from q_i and i . Substitution of these definitions gives equality of the corresponding OD classes, with or without the allowed V_λ parameters. Their hereditary subclasses are consequently equal. $\square$

The copying principle is already present in the supplied synthesis. Its role here is to show that the multiplicity appearing there costs no more consistency strength than the first rigid class. These copies do *not* answer the question whether there are distinct models N_q .

6.2 The definable-section obstruction already occurs in this model

Definition 6.2. A *complete section* is a set $T \subseteq D_\lambda$ meeting every $q \in Q_\lambda$. It is *thin* if every such intersection has cardinality below λ . A partial section need not meet every class; it is thin when all its nonempty fibres are smaller than λ .

Proposition 6.3 (Simultaneous full-width fibres). If q is rigid and q_i are its copies from Lemma 6.1, then for every $T \subseteq D_\lambda$ in OD_{V_λ} and every $i < \lambda$,

$$|T \cap q_i| \in \{0, \lambda\}.$$

The same λ classes serve all these sets T simultaneously. Every definable complete section therefore has λ full-sized fibres, and no thin complete section is in OD_{V_λ} .

Proof. $T \cap q_i$ is definable from q_i and allowed low-rank parameters. If nonempty, it is a family of short cofinal subsets of λ , so rigidity gives cardinality at least λ . It is a subset of q_i , whose cardinality is λ . No choice of q_i in this argument depends on T . $\square$

Proposition 6.4 (Small families do not evade the obstruction). If q is rigid and $\mathcal{T} \in \text{OD}_{V_\lambda}$ has $|\mathcal{T}| < \lambda$, with each $T \in \mathcal{T}$ a thin partial section, then $(\bigcup \mathcal{T}) \cap q = \emptyset$. Consequently no nonempty such family consists of complete sections, and no such family of partial sections has complete union.

Proof. For each $T \in \mathcal{T}$ meeting q , put $u_T = \bigcup (T \cap q)$. All members of q are countable, so $|u_T| \leq |T \cap q| \cdot \aleph_0 < \lambda$, and u_T is cofinal. The family of these u_T is definable from q and $\mathcal{T}$ and has size at most $|\mathcal{T}| < \lambda$. Rigidity says it must be empty. Notice that we do not take the union of the possibly unbounded sizes $|T \cap q|$ over all T ; the family of compressed cofinal sets is the counterexample directly. $\square$

These consequences were obtained from ultraexactingness in the dossier. The new point is that Theorem 5.5 and Lemma 6.1 realize all of them in an ordinary measurable-cardinal Prikry extension.

6.3 Proof of the equiconsistency theorem

Proof of Theorem 1.3. $\text{MR} \Rightarrow \text{RC}$ follows by taking $i = 0$. The implication $\text{RC} \Rightarrow \text{HR}$ is Corollary 2.4. Proposition 3.1 gives the measurable inner-model lower bound from HR, and hence the corresponding consistency implication.

For the other direction, begin with a model with a measurable cardinal, normalize a witnessing measure, and use ordinary Prikry forcing. Theorem 5.5 supplies a κ satisfying RS and a rigid class at κ . Lemma 6.1 upgrades this to MR. Thus the consistency of one measurable cardinal gives the consistency of the strongest of the three calibrated assertions. Together the two directions give all the stated equiconsistencies.

As usual, this is the formal relative-consistency argument using the forcing theorem and the core-model theorem within models of set theory. It does not add the assumption that there exists a transitive *set* model of either theory. Presenting the construction with transitive $W \subseteq W[G]$ is a convenient semantic notation, not a stronger hypothesis in the consistency statement. $\square$

Consequence for the inconsistency search

The existence of rigid classes, the 0-or- λ law for definable fibres, and the exclusion of small definable families of thin sections are compatible with ZFC at exactly measurable-cardinal consistency strength. Any contradiction obtained from this package alone would also refute the consistency of a measurable cardinal. To go beyond that calibration, an argument must use additional structure: for example the embedding dynamics, simultaneous preservation of predicates, or interaction with large cardinals below λ .

7 The canonical Boolean algebra of a class

We now extract a second result from rigidity. This construction is a specialization of Vopěnka's theorem. A modern statement with the single-parameter generality needed here is Goldberg's Theorem 4.8 [8]. We give the construction and the extension-equality argument, so no step is hidden in the word "generic."

Throughout this section, $q = [a] \in Q_\lambda$; rigidity will be invoked only where explicitly stated. Define, externally in V ,

$$\mathbb{A}_q = \{A \subseteq q : A \in \text{OD}_q\}.$$

This is a set of subsets of q , with the ordinary Boolean operations. It need not belong to H_q . Choose its canonical OD_q enumeration

$$f : \gamma \longrightarrow \mathbb{A}_q$$

by the least codes for ordinal definitions from q . Transfer the Boolean operations to the ordinal γ using f ; write $\mathbb{B}_q$ for the resulting coded algebra.

Lemma 7.1 (Internal completeness). $\mathbb{B}_q$ is a complete Boolean algebra in H_q . The decoding map f is used externally and is not asserted to belong to H_q .

Proof. The code for the algebra consists of relations and functions on ordinals which are ordinal definable from q . Their transitive closures consist of ordinal data and such definable graphs. They therefore belong to H_q . The Boolean-algebra identities are inherited from subsets of q .

For a family $I \subseteq \gamma$ belonging to H_q , the set I is itself OD_q . Hence

$$\bigcup \{f(\xi) : \xi \in I\}$$

is an OD_q subset of q , and has a code in $\mathbb{B}_q$. That code is the supremum of I . This verifies completeness for all families which *belong to* H_q , exactly as required. We are not claiming external completeness for arbitrary families in V . $\square$

Definition 7.2. For each $b \in q$, let

$$G_b = \{\xi < \gamma : b \in f(\xi)\}.$$

We view G_b as an ultrafilter on $\mathbb{B}_q$.

Lemma 7.3 (Every representative gives a generic filter). G_b is H_q -generic for $\mathbb{B}_q$ for every $b \in q$. Furthermore $b \in H_q[G_b]$ and $G_b \in M_b$.

Proof. Membership at a point respects intersections and complements, so G_b is an ultrafilter. Let $\mathcal{A} \in H_q$ be a maximal antichain of positive elements of $\mathbb{B}_q$. Its decoded sets are pairwise disjoint. Their union is q : otherwise the code for the nonempty complement would be a nonzero element incompatible with every member of $\mathcal{A}$, contradicting maximality. Thus exactly one decoded member contains b , and G_b meets $\mathcal{A}$. This proves genericity.

For $\alpha < \lambda$, let e_α be the code for $\{x \in q : \alpha \in x\}$. The sequence $\langle e_\alpha : \alpha < \lambda \rangle$ is ordinal coded and OD_q , so belongs to H_q . Then

$$b = \{\alpha < \lambda : e_\alpha \in G_b\},$$

which reconstructs b in the generic extension.

Finally, q is ordinal definable from b , as its finite-change class. Thus G_b is ordinal definable from b , and is a set of ordinals. It follows that $G_b \in M_b$. $\square$

7.1 Why the extension is exactly the representative HOD

Theorem 7.4 (Exact generic-extension identity). For every $b \in q$,

$$H_q[G_b] = \text{HOD}_{\{q,b\}} = M_b.$$

For any $a, b \in q$, $M_a = M_b$. We may consequently denote this common extension by M_q .

Proof. Because q is ordinal definable from b , adjoining q as an allowed parameter in addition to b does not change the hereditary class: $\text{HOD}_{\{q,b\}} = M_b$. Also $H_q \subseteq M_b$ and $G_b \in M_b$. The transitive model M_b evaluates all H_q -names, so $H_q[G_b] \subseteq M_b$.

For completeness, we prove the reverse inclusion by explicitly coding hereditarily definable sets. There is a fixed definable enumeration scheme $F_z(\xi)$, $\xi \in \text{Ord}$, of the sets ordinal definable from q, z . To obtain it, well-order the rank-local definition codes and use $\emptyset$ for failed uniqueness codes. Every $\text{OD}_{q,z}$ set occurs. The evaluation of each code is a first-order set-theoretic operation; no global truth predicate is involved.

Fix $x \in M_b$. Choose an ordinal η large enough that every member of $\text{tc}(\{x\})$ occurs among $F_b(\xi)$ for $\xi < \eta$. Such a bound exists because the transitive closure is a set and every one of its elements is ordinal definable from b . Increase η if necessary to include a code ξ_x for x itself.

For each $z \in q$ define

$$D_z = \{\xi < \eta : \text{tc}(\{F_z(\xi)\}) \subseteq \text{ran}(F_z \upharpoonright \eta)\}.$$

The image $F_z''D_z$ is transitive. Indeed, if a value has its full transitive closure represented below η , so does each of its members. Keep only the least index for each repeated value, obtaining $D_z^* \subseteq D_z$. Put a relation on this set by

$$\xi E_z \nu \iff F_z(\xi) \in F_z(\nu).$$

This is a well-founded extensional relation isomorphic to the transitive set $F_z''D_z$.

For each $\xi, \nu < \eta$, the subsets of q expressing $\xi \in D_z^*$ and $\xi E_z \nu$ are OD_q . Their Boolean codes, indexed by η and $\eta \times \eta$, form sets of ordinal data in H_q . Evaluating them with G_b therefore constructs D_b^* and E_b inside $H_q[G_b]$. This relation is well-founded in V , hence also in the inner model $H_q[G_b]$, and its Mostowski collapse is absolute. The collapse at the least index representing x is exactly x . That index is an ordinal, so it is available in $H_q[G_b]$. Thus $x \in H_q[G_b]$.

Finally, if $a =^* b$, the finite set $a \triangle b$ is ordinal definable using its finitely many ordinal elements. Each of a and b is therefore ordinal definable from the other. Substitution of definitions, including those for all elements of the transitive closure, gives $M_a = M_b$. $\square$

The identity is stronger than merely observing that the representatives are generic over some inner model. It identifies a *single* forcing algebra, a *single* ground, and a *single* extension shared by all representatives of the class.

8 Homogeneity, singularization, and rank agreement

8.1 Finite changes give an internal automorphism action

Let $\Gamma = ([\lambda]^{<\omega}, \Delta)$. This is an ordinal-definable group and belongs to H_q . For $d \in \Gamma$ define

$$\rho_d(A) = \{x \triangle d : x \in A\} \quad (A \in \mathbb{A}_q).$$

Transporting ρ_d through f gives an automorphism of $\mathbb{B}_q$. The entire coded action is in H_q .

Lemma 8.1 (Weak homogeneity). The action of Γ makes $\mathbb{B}_q$ weakly homogeneous in H_q : for any nonzero $p, r \in \mathbb{B}_q$, some $d \in \Gamma$ has $\rho_d(p) \wedge r \neq 0$. Moreover

$$\rho_d[G_b] = G_{b \triangle d}.$$

Proof. Choose $x \in f(p)$ and $y \in f(r)$. Both lie in q , so $d = x \triangle y$ is finite. The point y lies in $\rho_d(f(p)) \cap f(r)$, proving compatibility. The witness d is an element of the internal group Γ , so this establishes precisely the asserted statement of H_q . The equality of filters follows by applying the equivalence $b \in A \iff b \triangle d \in \rho_d(A)$ to each Boolean code. $\square$

Lemma 8.2 (Atomlessness under rigidity). If q is rigid, $\mathbb{B}_q$ is atomless in H_q .

Proof. A nonzero code decodes to a nonempty OD_q subset A of q . Since $A \subseteq C_\lambda$, rigidity gives $|A| \geq \lambda$. Choose distinct $x, y \in A$ and an ordinal $\alpha \in x \triangle y$. Intersecting A with $\{z \in q : \alpha \in z\}$ and with its complement splits its code into two nonzero elements. All splitting codes are in H_q . Thus no nonzero element is an atom. $\square$

Theorem 8.3 (Canonical singularization attached to rigidity). Suppose $\text{RS}(\lambda)$ and $\text{Rig}(\lambda, q)$. There is a complete Boolean algebra $\mathbb{B}_q \in H_q$ such that:

1. $\mathbb{B}_q$ is atomless and weakly homogeneous, with the finite-change group acting as above.
2. Every $b \in q$ yields an H_q -generic G_b , and $H_q[G_b] = M_q$ independently of b .
3. H_q regards λ as inaccessible; M_q regards it as a strong limit cardinal of cofinality ω . In particular $H_q \neq M_q$.
4. H_q satisfies that $\mathbb{B}_q$ forces λ to remain a cardinal and to have cofinality ω . The algebra does not have the λ -chain condition in H_q .

If additionally $V_\lambda \subseteq \text{HOD}^V$, then $H_q = N_q$,

$$V_\lambda^{H_q} = V_\lambda^{M_q} = V_\lambda^V,$$

and $\mathbb{B}_q$ forces that no bounded subsets of λ are added.

Proof. The first two clauses are Lemmas 7.1, 7.3, 8.1, 8.2 and Theorem 7.4. Corollary 2.4 gives regularity in H_q . The ordinal λ is a cardinal of each inner model, since any inner-model collapse would be an ambient collapse.

Strong-limitness is downward absolute here. If an inner model of Choice had an injection of λ into its power set of some $\mu < \lambda$, that would be an ambient injection into $\mathcal{P}^V(\mu)$, contradicting the ambient strong-limit property. Thus λ is strong limit in H_q and in M_q . It is inaccessible in the former. In the latter any $b \in q$ supplies its cofinal ω -enumeration. That enumeration cannot belong to H_q , so the extension is proper.

In a weakly homogeneous Boolean algebra, the Boolean value of a sentence with check parameters from the ground is either zero or one. Indeed, if nonzero conditions forced opposite answers, an automorphism could make their images compatible, while fixing those check parameters. The assertion that λ is a cardinal of cofinality ω holds in the actual extension M_q . The truth lemma and weak homogeneity therefore imply that it is forced by the top condition in H_q .

If $\mathbb{B}_q$ had the λ -chain condition, take a name for a cofinal map $\omega \rightarrow \lambda$. For each n , a maximal antichain deciding its n th value has size below λ . Regularity of λ in H_q then bounds the union of all possible values for all $n < \omega$ below λ , contradicting cofinality. Hence the chain condition fails.

Now assume $V_\lambda \subseteq \text{HOD}^V$. Each low-rank parameter can be replaced in a definition by an ordinal definition. Consequently $\text{OD}_{V_\lambda \cup \{q\}} = \text{OD}_q$, so $N_q = H_q$. Both H_q and M_q contain the full ambient V_λ and are contained in V , proving the rank equality. In particular the actual extension has no new bounded subsets of λ .

To obtain the forcing assertion rather than just a statement about this one generic extension, let $X = (V_\lambda)^{H_q} \in H_q$. The sentence that every subset of every $\alpha < \lambda$ belongs to $\check{X}$ uses only ground check parameters. It holds in M_q , so the preceding weak-homogeneity argument makes its Boolean value one. $\square$

Three distinctions that must not be suppressed

First, H_q is a ground of M_q , not necessarily a ground of the whole ambient universe V . Second, preserving λ as a cardinal does not assert preservation of every cardinal above it. Third, adding no bounded subsets of λ does *not* mean ω -distributivity: the extension adds a new countable cofinal sequence of ordinals below λ .

There is also no forcing assertion of the form $\dot{b} \in \check{q}$ here. For a rigid class, $q \notin H_q$, so that is not a legitimate use of a ground-model check name. The class is used externally to construct the ordinal-coded algebra; its actual representatives then induce the generic filters.

8.2 The representatives cannot be mutually generic

Proposition 8.4. If q is rigid and $a, b \in q$, the filters G_a and G_b are not mutually generic over H_q for $\mathbb{B}_q \times \mathbb{B}_q$. There are exactly λ distinct point filters G_b , $b \in q$, and all of them generate M_q .

Proof. Different points give different filters by the membership codes e_α , so there are λ of them. Their extensions coincide by Theorem 7.4. They are proper because Corollary 2.4 makes λ regular in H_q , whereas each $b \in q$ gives a cofinal ω -sequence in M_q .

Recall the intersection fact for mutually generic extensions: if G, H are mutually generic over a transitive ground N , then $N[G] \cap N[H] = N$. Here is the relevant proof. For a common subset x of an ordinal, take names σ, τ from the two coordinates and a product condition (p, r) forcing their equality. If two extensions of p forced opposite answers to $\alpha \in \sigma$, first extend r to decide $\alpha \in \tau$. One of the two resulting product conditions would contradict the forced equality. Thus p decides every membership question, making x a ground-model set. For an arbitrary common set outside N , choose one of least rank. Its members are all in N ; a ground well-order of the appropriate ground rank converts it to a common set of ordinals outside N , contradicting the case just proved.

Mutual genericity of G_a, G_b would consequently make their common extension equal to H_q , contrary to properness. $\square$

This does not say that the algebra has no product-generic filters in a further universe. It says that the representatives of this one ambient finite-change class cannot form such a pair.

9 What the result does and does not settle

The consistency theorem identifies a precise boundary inside the supplied research program. Rigidity has exactly measurable consistency strength, and its definable-section consequences already occur in the same construction. The ultraexacting embedding hypothesis is not needed to realize them, and its additional force must be sought in properties not used by the Prikry construction.

Theorem 8.3 adds a second boundary. Under $V_\lambda \subseteq \text{HOD}$, the “regular inside, singular outside” picture from the synthesis is realized by an actual weakly homogeneous set forcing *between the two indicated hereditary models*. This converts a cofinality discrepancy into a specific forcing decomposition. It neither identifies that forcing with ordinary Prikry forcing nor derives a normal measure in H_q . The measurable lower bound is an inner-model conclusion, not a classification of $\mathbb{B}_q$.

9.1 Remaining questions with sharper targets

Question 9.1. Must there be distinct rigid q, r with $N_q \neq N_r$ under an ultraexacting hypothesis? The measurable construction and the copying lemma deliberately give many classes with the *same* hereditary model, so multiplicity of classes alone cannot answer this.

Question 9.2. What extra property of $\mathbb{B}_q$, arising from an ultraexacting embedding rather than from rigidity alone, distinguishes it from the measurable Prikry examples? Potential properties must involve the embedding action or a simultaneous family of predicates; atomlessness, weak homogeneity, and singularization by themselves are insufficient.

Question 9.3. Does ordinary exactingness imply the existence of a rigid class, or the nonexistence of a definable thin complete section, in full generality? The consistency calibration does not prove this implication. The quotient-preserving cone maps used in the upper bound belong to a particular forcing construction, not to every exacting witness.

We make no preservation claim for a strongly compact cardinal below the Prikry point, no new consistency claim for cover-exactingness above strong compactness, and no determination of the successor-collapse alternative from the synthesis. Those are different targets and require additional arguments. The published consistency relationship between ultraexactingness and I0 likewise remains a separate result [10].

10 Proof audit and dependency ledger

Conclusion	Logical dependencies
Rigidity $\Leftrightarrow$ regularity of N_q	Definition of rigidity; least-order-type compression; absoluteness of ordinal order type. No large-cardinal input.
Measurable lower bound	Classical Dodd–Jensen covering and its canonical core-model well-order; the application in Proposition 3.1.
Measurable upper bound	A normal measure; the Prikry property and rank preservation proved here; cone isomorphisms preserving the orbit name.
λ simultaneous rigid copies	Definable ordinal pairing and substitutions of ordinal definitions. This copying mechanism is also in the dossier.
Exact forcing decomposition	Ordinal-coded Vopěnka algebra; maximal-antichain genericity; the explicit collapse-of-definition-codes proof.
Weak homogeneity and atomlessness	Finite symmetric differences; rigidity; ordinal membership tests.
No bounded subsets in the canonical extension	The <i>additional</i> assumption $V_\lambda \subseteq \text{HOD}^V$, followed by weak homogeneity. It is not inferred from rigidity alone.

10.1 Checks against common proof failures

The cardinal bounds are ambient bounds until a model superscript is given. When transferring smallness to an inner model, the proof uses the *absolute order type* of a subset of λ , not an unjustified assumption of cardinal correctness. The compression proof does not union an unbounded collection of sizes below a singular cardinal. The Prikry homogeneity argument fixes the class name and all ground check parameters, not the generic sequence. The Boolean algebra is complete in H_q , not necessarily in V . Its ordinal code belongs to H_q , while its decoding map and the class q need not. Finally, the assertion about all generics adding no bounded subsets is deduced from weak homogeneity and ground check parameters, not from the behavior of one extension alone.

10.2 Verification status

All newly used deductions are written as mathematical proofs in this report. The supplied Lean files were reference material for the definitions and the scope of the earlier claims; the present equiconsistency argument and the canonical forcing analysis have not been formalized in Lean. The Dodd–Jensen theorem is an imported classical theorem. The report does not claim an independent formal check or a new proof of that theorem. Bibliographical verification establishes the cited classical inputs, not global novelty of the particular formulation proved here.

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